

Amit Rahman

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WORK AUTHORIZATION

Singapore Citizen; eligible for H-1B1 US visa, no lottery/petition required

EDUCATION

National University of Singapore (NUS) May 2025
Bachelor of Engineering (Honours) in Computer Engineering - 4.1 / 5.0 GPA
• **NUS Engineering Scholar** - Awarded to 50 engineering students in faculty cohort of 1500

EXPERIENCE

Qube Research & Technologies Aug 2025 – Present
Software Engineer - Core trading systems [C++20, CMake, Linux (RHEL 7-9), Windows, Python, Bash] Hong Kong SAR
• Development owner of the firm’s C++ margin service computing real-time initial margin (SPAN, PRiMe, VaR) for global derivatives and cash-equity portfolios over an event-driven pub/sub mesh, with client libraries in C++, Python and C#
• Redesigned the concurrency model for throughput, sharding contended shared state and moving computation onto an asynchronous, pooled pipeline offloading heavy requests run from the critical path.
• Architected plugin framework for margin calculation engine, to be extensible with runtime configuration for quick build out, growing the service to 3 new exchanges and asset classes that could be independently operated as micro-services
• Designed the what-if margining API so the CRO and trading desks can price hypothetical portfolios against live positions, informing capital deployment and broker dispute decisions
• Owns the service’s automated testing, reproducing a live multi-host trading environment deterministically and checking margin against known-good baselines, turning a manual test plan into continuous regression coverage
• Modernized monorepo’s Linux and Windows CI onto a shared build configuration with better caching and incremental builds, leveraging CMake presets, CMake driven python packaging, compiler caching set up across development and CI environments; cache-hot rebuilds dropping times ~30 - 60% across the board

Software Engineer Intern - Core trading systems May 2025 - Aug 2025
• Eliminated a memory-allocation bottleneck on the FIX hot path, cutting allocation CPU from 24% to 6% with pooled allocation and fewer page faults by integrating a C++17 PMR-wrapped pool allocator with logarithmic bucket sizing, cutting p99 tail latency by 4x
• Introduced lightweight service instrumentation and microbenchmarking suite, enabling continuous telemetry collection
• Improved parsing performance with cache-friendlier data structures (2 to 9x faster lookups, -68% more inbound throughput), supported by a profiling and benchmarking harness

Grasshopper Sep 2024 – Feb 2025
Software Engineer Intern [Rust] Singapore
• Authored framework for declaratively creating custom and performant Wireshark protocol dissectors written in Rust
• Spearheading new developments of a high-performance Rust implementation parser generator for Google’s Protobuf, enhancing data serialization efficiency
• Conducting thorough performance benchmarking ensuring Protobuf parsers meets stringent high-frequency trading requirements

Kisi Inc. Feb 2024 – Aug 2024
Software Engineer I - Embedded systems [C, CMake, Python] New York
• Developed critical firmware components including a thread-safe networking library and encryption module, resulting in an 80% reduction in networking latency and a 100x improvement in encryption/decryption speed in native C code.

HydraX May 2023 – Aug 2023
Software Engineer Intern - Cloud platform team [Python] Singapore
• Migrated legacy AWS resources & CI/CD to infrastructure-as-code, for private digital exchanges

SKILLS

Languages: Modern C++ 20, C, Rust, Python, Bash
Technologies: Docker, Kubernetes, AWS, GCP, FreeRTOS, CMake, Linux, Perf, Git, Grafana, Vim
Coursework focus areas: Systems Programming, Compilers, Programming Languages